

Hotspots

In some parts of the world, plates of oceanic crust are moving slowly over extra-hot parts of the mantle called hotspots. Each hotspot creates a volcano in the moving crust. Once the volcano moves off the hotspot, it becomes extinct, and a new volcano erupts in its place. Over millions of years, this process creates a chain of islands.

Hawaii

Longest island chain

Location Central Pacific

Highest point 13,796 ft (4,205 m)

Last eruption Constant activity

The Hawaiian islands were formed by a hotspot in the middle of the Pacific plate, which is moving northwest over the hotspot at the rate of 3.5 in (9 cm) a year. For more than 80 million years, the heat has been making volcanoes erupt through the moving plate. This has formed a chain of islands and submerged seamounts stretching about 3,700 miles (6,000 km) across the Pacific Ocean.

► FIRE FOUNTAIN

The Hawaiian hotspot now lies beneath the most southerly island of Hawaii, where the most active volcano on Earth, Kilauea, has been erupting almost continuously since 1983.



Iceland

Spreading rift

Location North Atlantic

Highest point 6,923 ft (2,110 m)

Last eruption Constant activity

Iceland is a vast mass of basalt lava that has erupted from a hotspot lying below the Mid-Atlantic Ridge. The plates of the crust are moving apart beneath the island, causing the eruption of many volcanoes and geysers. But since the hotspot is not under a moving plate, it has not created an island chain.



Réunion

Tropical hotspot

Location Western Indian Ocean

Highest point 10,072 ft (3,070 m)

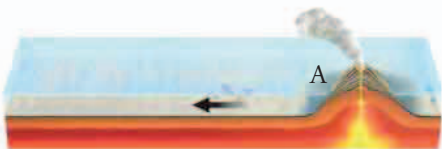
Last eruption Constant activity

Réunion is at the southern end of a short chain of volcanic islands in the Indian Ocean that also include Mauritius. The chain extends further north underwater, and underlies the coral islands of the Maldives. The hotspot now lies under the southeast corner of Réunion, causing the regular eruption of the Piton de la Fournaise volcano.

VOLCANIC CHAINS

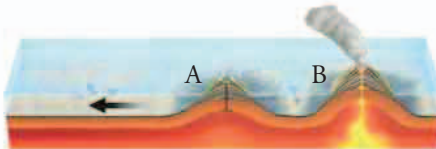
A hotspot beneath the crust heats the rock so that it expands and rises. Some rock melts and erupts as basalt lava, forming a volcanic island. When plate movement carries the volcano off the hotspot,

it stops erupting. As the rock beneath it cools and contracts, the island sinks and eventually becomes a submerged seamount. Some hotspots have created hundreds of these volcanic islands and seamounts.



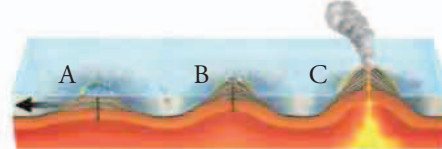
▲ 1. ERUPTION

A stationary hotspot burns a hole through the Earth's moving crust. This creates a volcano (A), which erupts and forms an island.



▲ 2. EXTINCTION

Over millions of years, the island moves off the hotspot and starts sinking, while a new volcano (B) erupts.



▲ 3. SUBSIDENCE

As the first island in the chain sinks below the waves, the second moves off the hotspot, and a third volcano (C) erupts.